Shreyan Balaji Nalwad

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EDUCATION

Carnegie Mellon University — Pittsburgh, PA
Master of Science in Computational Biology — May 2027

Vellore Institute of Technology — Vellore, India
Bachelor of Technology in Computer Science and Engineering with specialization in Bioinformatics — May 2025

EXPERIENCE

Zhao Biophotonics Lab — Pittsburgh, PA
Graduate Research Assistant — Jan 2026 - Present

- Designed Denoising Diffusion Probabilistic Model (DDPM) for 'channel-to-channel generation' task, enabling cost-effective virtual fluorescence microscopy by synthesizing multiplexed tissue stains without expensive physical reagents.
- Engineered end-to-end HDF5 data pipeline and stabilized diffusion process by integrating FiLM layers into custom 8M-parameter architecture, driving generative model to convergence.

Vyuhaa Med Data — Hyderabad, India
AI Integration Engineer - Intern — Oct 2024 - July 2025

- Architected 3-stage inference engine (object detection, classification, segmentation) facilitating downstream morphometric analysis, starting with abnormal cell detection followed by targeted patch-centering and pruning for high-confidence regions.
- Optimized edge AI performance latency from 500s to 80s by parallelization and eliminating redundant OpenSlide handle initializations and threading, deploying on Jetson Orin Nano.
- Formalized active learning feedback loop by isolating negative-sample tissue artifacts, collaborating with in-house pathologist as oracle to automate generation of new training batches directly from expert annotations.
- Initiated and drove pipeline migration from YOLOv7 to YOLOv11, streamlining deep learning workflow, yielding 90% precision and 10x boost in recall for cervical cancer detection on clinical-grade WSI.

PROJECT

Mitograph: Mitochondrial Knowledge Graph for VUS Pathogenicity Prediction — Feb 2026

- Architected a heterogeneous Knowledge Graph and Graph Attention Network (GATv2Conv) to predict pathogenicity for 1,228 mitochondrial Variants of Uncertain Significance (VUS) across 808 disease phenotypes, attaining a 0.830 Test AUPRC.
- Structured biologically grounded node representations via LLM sentence-transformers and circular positional encoding, ensuring zero data leakage while extracting GNN attention weights to deliver transparent biological interpretability.

MUFFLE: Multimodal Framework for Federated Learning — Jan 2026

- Spearheaded multimodal fusion strategy for a 11-person team using NVFlare, standardizing data retrieval from AWS S3 and leading technical design consultations securing the "**Best Collaboration Award**".
- Orchestrated integration of WSI and RNA-seq encoders to stratify patients into 3 distinct risk clusters. Enhanced interpretability leveraging gated attention fusion mechanism, enabling visual correlation of tumor morphology.

SPARTA: Spatial Metabolite Alignment and Ratio Temperament Analysis — Dec 2025

- Built a Streamlit tool for MALDI-MSI analysis, utilizing intensity-weighted centroid alignment to superimpose metabolic maps while preserving spatial variance.
- Programmed SNR-floor filtering and log-transformed co-localization logic (log2), identifying metabolic fronts and tumor boundaries across 100+ METASPACE-annotated metabolites.

Hybrid Sequential Architecture of CNN and Transformer for Diabetic Retinopathy Grading — Oct 2024

- Published and presented **first-author** research (Taylor & Francis) at Com-IT-Con (Oct 2024), authoring core technical framework sequentially coupling CNNs and Vision Transformers (ViT) to reach 87% grading accuracy.

SKILLS

Machine Learning & AI: PyTorch, CUDA, Generative Models (DDPM, VAE), Computer Vision (ViT, YOLO), Graph ML (GNNs).
Spatial & Molecular Omics: MALDI-MSI (pyimzML), WSI (OpenSlide, Napari), scRNA-seq (Seurat, Bioconductor), METASPACE.
MLOps & Cloud Infrastructure: AWS (Certified Cloud Practitioner), S3, Brev.dev, NVIDIA Jetson, Edge Inference, Git.
Languages: Python, Go, R, C++, Java, SQL.